Adityanarayanan Buddharajan

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Summary

Full-stack Software Engineer with 2 years of experience at DTCC, building fintech platforms using Java, Spring Boot, and Angular, and shipping features end-to-end from design through production. Recently built the real-time streaming pipeline for a Microsoft-collaboration sports analytics platform on Azure, working across the stack under ambiguity to ship incrementally. Comfortable integrating AI-assisted development tools (eg: Copilot/Claude code) into daily workflow to improve velocity. Seeking Full-stack / Software Engineer roles.

Technical Profile

Languages: Java, JavaScript, Python, SQL
Backend: Spring Boot, RESTful APIs, Microservices, Oracle Database, PostgreSQL, DynamoDB
Frontend: React, Angular, Next.js, GraphQL, Tailwind CSS, HTML, CSS
Cloud & DevOps: AWS, Azure (Functions, Event Hub, Container Apps), Docker, Kubernetes (fundamentals), Jenkins, CI/CD
AI-Assisted Development: GitHub Copilot, Claude Code
Practices: Agile/Scrum, System Design, Data Structures & Algorithms, Unit Testing, TDD, Code Reviews

Work Experience

Depository Trust and Clearing Corporation (DTCC) **July 2023 – July 2025**
Software Engineer (Software Analyst → Senior Software Analyst) *Chennai, India*

- Built a CSV upload and processing dashboard (Angular, Spring Boot) with a **scheduled async batch pipeline** that triggered every 5 minutes to validate and process files up to 5MB (˜1,000 records), enforcing field-level rules (data type checks, date validation, nullable/required constraints), cutting operations teams' manual effort by 80%.
- Designed an Ag-Grid dashboard for Corporate Actions, consolidating announcement lifecycle stages into a single real-time view via WebSocket-based updates from backend to frontend, with optimized pagination and filtering to keep large batch-generated announcement volumes clean and navigable.
- Replaced a full-list SQL query pattern that returned all announcements on every call with paginated, on-demand queries triggered per user action (e.g., "next" clicks), and added indexes on unique identifiers, **reducing file processing time by 40%**.
- Designed and documented RESTful APIs using Swagger/OpenAPI for contract-first microservices communication between the Angular frontend and Spring Boot backend, streamlining cross-team integration.
- Implemented resource-level **Role-based Authentication** and **Authorization** using Auth Guards and session state management, dynamically restricting record visibility and views based on user role and login credentials.
- Introduced unit and integration testing with JUnit 5 and mocking frameworks, complemented by manual UI testing, catching issues earlier and reduced production defects.

Projects

SportScore: Real-Time Multi-Sport Tracking Platform | Microsoft Collaboration | Live Demo **2026**

- Owned the real-time streaming pipeline, building an event-driven architecture using Azure Event Hub and Azure Functions to process more than 2,000 live sports events at scale.
- Built "Radio Mode," a live audio commentary feature using **Azure Neural TTS** and **SignalR**, delivering real-time broadcast-style updates to users.
- Extended event detection to baseball and F1, designing sport-specific event handlers on top of a shared detection framework.
- Diagnosed and resolved backend scaling and latency issues in production, replacing declarative retry annotations with a manual retry loop for greater control over failure handling, and fixing a Redis caching outage through a redesigned **fail-open pattern** improving pipeline reliability under live-match load.
- Designed and built a Redis caching layer with a fail-open pattern and **tiered TTL strategy** (6-hour/30-second split) to balance data freshness against ESPN API rate limits, later migrating the cache to Upstash for improved reliability.

Real-time Speech Emotion Detection using AI | *Python, LSTM, Streamlit* **2023**

- Developed a deep learning model using **LSTM networks** to classify human emotions from speech audio signals, extracting MFCC and spectral features for improved accuracy.
- Built a complete machine learning pipeline covering audio preprocessing, feature engineering, training, and evaluation, with hyperparameter tuning to optimize prediction performance.
- Built and deployed a Streamlit web application for real-time inference.

Education

University College Dublin **Sept 2025 – Sept 2026**
MSc in Computer Science (GPA: 3.80/4.2) *Dublin, Ireland*

Madras Institute of Technology, Anna University **Aug 2019 – June 2023**
B.Tech in Information Technology *Chennai, India*